

A standardised clinician-supervised digital workflow for sub-symptom-threshold aerobic exercise after concussion

Zac Lewis (ORCID 0009-0002-4267-0451) - Concussion Education Australia

A standardised clinician-supervised digital workflow for sub-symptom-threshold aerobic exercise after concussion

Author: Zac Lewis, Registered Osteopath (AHPRA-registered), B.Clin.Sci, M.Ost.Med

Affiliation: Concussion Education Australia Pty Ltd, 2 Wordsworth St, Byron Bay NSW 2481, Australia **ORCID:** 0009-0002-4267-0451 **License:** CC-BY 4.0

Abstract

The 2022 Amsterdam international consensus made sub-symptom-threshold aerobic exercise (SSTAE), prescribed from an individually measured heart-rate threshold (HRT) derived by a graded exercise test, a first-line treatment for concussion. Translating that prescription into practice requires a graded test, an individualised heart-rate band, monitored between-visit sessions, and progression governed by heart rate rather than session duration. This protocol specifies a standardised five-stage clinician-supervised workflow for delivering SSTAE with digital monitoring. The clinician interprets the test, sets the band, approves each progression, and makes all clearance decisions; the tool executes, paces, monitors and records. Two distinct symptom thresholds are used and must not be conflated: a rise of 3 points or more defines the diagnostic threshold during graded testing, and a rise of 2 points or more is the within-session training stop rule.

Guidelines

Intended users. Clinicians supervising concussion rehabilitation, and the patients they prescribe for. This workflow is clinician-directed throughout: it informs, it never decides.

Two symptom thresholds - do not conflate.

- **3 points or more** above resting baseline = the **diagnostic** provocation criterion during the graded test (Stage 2).
- **2 points or more** above pre-session score = the **within-session training stop rule** during home sessions (Stages 3-4).

Reduce, don't rest. After a symptom flare the response is to continue daily sub-symptom exercise at *reduced* intensity, not to prescribe a rest day. The only prohibition is sustained supra-threshold exercise.

Progression is by heart rate, not duration. Sessions remain ~20 minutes throughout. The ceiling rises; the session length does not.

Real signal or no signal. Heart rate is displayed only when actually measured. The workflow pauses on "no reading" rather than acting on an estimate. Only live-verified wearable sessions may advance the band.

Safety warnings

SCAT6 emergency red flags - halt and refer immediately at any stage. Severe or worsening headache; repeated vomiting; seizure; focal weakness or numbness; slurred speech; increasing confusion or drowsiness; loss of consciousness. Any positive finding blocks testing and routes to urgent medical review. No HRt is produced from an unsafe stop.

Do-not-exceed ceiling. The upper bound of the prescribed band is an explicit do-not-exceed ceiling. An advance never exceeds the measured HRt.

Scope limit. This workflow supplies objective aerobic-tolerance evidence. It does not, by itself, clear an athlete for contact. Clearance and return-to-sport remain clinician-led decisions under the consensus graded strategy.

Materials

- Treadmill (or cycle ergometer where a standardised alternative protocol is used)
 - Heart-rate monitor broadcasting the standard Bluetooth Heart Rate profile
 - Borg rate of perceived exertion scale (6-20)
 - 0-10 symptom/visual analogue scale
 - SCAT6/PCSS 22-item symptom inventory
 - SCAT6 emergency red-flag list
-

Procedure

Stage 1: Baseline and safety screen

Step 1. Record the symptom profile (Stage 1)

The patient selects the post-concussion symptoms they actually experience from the standard 22-item SCAT6/PCSS inventory, so that subsequent prompts ask only about relevant symptoms.

Step 2. Screen against the red-flag gate (Stage 1)

SAFETY - perform before any exertion. Screen the patient, with clinician oversight, against the SCAT6 emergency red-flag list: severe or worsening headache, repeated vomiting, seizure, focal weakness or numbness, slurred speech, increasing confusion or drowsiness, loss of consciousness.

Stopping criterion: any positive finding blocks the test and routes to urgent medical review.

Step 3. Capture the resting symptom score (Stage 1)

Record a 0-10 baseline immediately pre-test. This is the reference value for the provocation criterion in Stage 2.

Step 4. Pair the heart-rate source (Stage 1)

The patient pairs a Bluetooth wearable, enables camera photoplethysmography (resting spot-check only - motion artefact makes it unreliable during exercise), or the clinician selects manual entry.

Note. Apple Watch and Fitbit do not broadcast standard BLE heart rate; use manual entry for these devices.

Role: clinician confirms appropriateness and supervises; patient executes.

Stage 2: Graded threshold test to a measured HRt

A standardised graded ramp modelled on the Buffalo Concussion Treadmill Test (BCTT), built on the cardiac Balke protocol.

Step 5. Set the starting workload (Stage 2)

Treadmill at a fixed walking speed set by height: **3.2 mph** for patients 5 ft 10 in or under, **3.6 mph** for patients over 5 ft 10 in. Hold **0 degrees incline for the first minute**.

Step 6. Ramp the workload each minute (Stage 2)

Increase incline by **1 degree per minute** - degrees, not per cent - to a maximum of **15 degrees**. Once 15 degrees is reached, hold the incline and increase **speed by 0.4 mph per minute**.

Step 7. Monitor every minute (Stage 2)

Record heart rate, rate of perceived exertion (Borg 6-20), and a 0-10 symptom/VAS score **every minute**.

Step 8. Identify the heart-rate threshold (HRt) (Stage 2)

HRt is the heart rate at the first stage at which the symptom score rises **3 points or more above the resting baseline** - the validated diagnostic provocation criterion.

Do not conflate the 3-point diagnostic threshold with the 2-point within-session training stop rule used in Stages 3-4.

Step 9. Terminate the test on any termination criterion (Stage 2)

Terminate on any of:

- a **VAS rise of 3 points or more** (symptom-limited -> HRt obtained)
- **voluntary exhaustion (RPE above 17)** without symptom provocation
- reaching **90% or more of age-predicted maximum heart rate** (220 minus age)
- rapid clinical deterioration
- patient request

Step 10. Interpret the termination (Stage 2)

Three terminations carry three distinct meanings:

- **Symptom-limited** -> HRt obtained; interpretation **physiologic** (exercise intolerance present). Proceed to prescription.
- **Voluntary exhaustion without provocation** (RPE above 17, no rise of 3 points or more) -> interpretation **no-intolerance**: symptoms unlikely to be exercise-driven. Redirect the workup (cervicogenic / vestibular / mood). On a *re-test*, this is the recovery/clearance signal.
- **Red flag** -> halt and refer. No HRt is produced from an unsafe stop.

Role: clinician interprets the termination.

Step 11. Cool down (Stage 2)

2.0 mph, 0 degrees incline, 2 minutes.

Step 12. Record the prognostic flag (Stage 2)

At presentation, an absolute **HRT below 135 bpm** together with a blunted heart-rate response (**delta-HR (threshold minus resting) of 50 bpm or less**) predicts prolonged (over 30-day) recovery, with approximately **73% sensitivity and 78% specificity**.

This is recorded and surfaced to the clinician as a prognostic signal only. It does not alter the prescribed dose.

Output: HRt (bpm) plus interpretation, recorded as the first point on the recovery curve.

Stage 3: Individualised band prescription

Step 13. Set the intensity band (Stage 3)

Train at **80-90% of HRT**. The 80% floor was the intensity used in the pivotal randomised controlled trial; the 80-90% band is the operational range in the practice literature.

SAFETY: the upper bound is an explicit do-not-exceed ceiling. The lower bound guarantees a therapeutic stimulus.

Step 14. Set the duration (Stage 3)

Approximately 20 minutes per day, fixed from day one. Session length is **not** ramped up over the programme.

Common point of error: progression is by heart rate, not by lengthening the session. See Stage 4.

Step 15. Set the frequency (Stage 3)

Daily in the randomised controlled trials; **5-7 days per week** (“most days”) clinically. There are **no scheduled rest days**.

Step 16. Encode the within-session stop rule (Stage 3)

Stop the session if symptoms rise **2 points or more above the pre-session score** on the 0-10 scale. This 2-point *training* threshold is deliberately lower than, and distinct from, the 3-point *diagnostic* threshold of Stage 2.

Note on other populations. Conservative presets exist for moderate/severe traumatic brain injury and other neurological conditions (wider safety margin, lower and narrower band, shorter sessions). The safety logic is identical; only the dose defaults change. These presets are **not** validated by the concussion randomised controlled trials.

Output: a plain-language prescription the patient can follow, plus a structured band (low/high bpm) enforced live during sessions.

Stage 4: Live-monitored between-visit rehabilitation

Step 17. Deliver live band feedback during each home session (Stage 4)

Display the measured heart rate against the prescribed band in real time as *under / in-zone / over*, so the patient can self-dose to an otherwise invisible ceiling.

Step 18. Apply the within-session stop rule (Stage 4)

If symptoms rise 2 points or more above the pre-session score, prompt the patient to stop. This encodes the consensus tolerance band rather than relying on recalled verbal instruction.

Step 19. Log the session (Stage 4)

Record average and peak heart rate, completed minutes, pre-session and peak symptom scores, and a next-day flare flag.

Step 20. Return data to the prescriber (Stage 4)

Where the patient has entered a clinic code, every threshold test and training session syncs to the clinician record: HRt trajectory, band, minutes, symptom deltas, flares. Self-guided patients without a clinic code transmit nothing.

Step 21. Progress by heart rate - clinician-reviewable (Stage 4)

Progression in the randomised controlled trials is by **heart rate**: raise the target ceiling by approximately **+5-10 bpm** when the patient tolerates the current dose without provocation, while the session remains at ~20 minutes. Propose *advance* / *hold* / *regress* / *refer* / *retest* from recent session history for clinician review:

- **Advance** the ceiling (+5-10 bpm) after a run of clean **live-verified** sessions - no within-session provocation, no next-day flare. Only sessions whose heart rate was live-verified against a broadcasting wearable count toward the clean run. Manual or camera-PPG sessions never advance the band.
- **Regress - reduce, don't rest.** On recent repeated provocation, reduce the ceiling and continue daily sub-symptom exercise at the reduced intensity. The response to a flare is a lower dose, not a rest day. Regression is **never** gated: flare and safety evidence counts from every session, verified or not.
- **Cap at the measured threshold.** An advance never exceeds the measured HRt. Once the band reaches it, prompt a **re-test** rather than ratcheting higher - a fresh graded test is the only safe way to raise the band.
- **Hold** otherwise.

Apply a recency window to regression so that an old, resolved flare does not ratchet the ceiling down indefinitely once the patient is running clean.

Note. The requirement for a *run* of verified clean sessions, and a minimum-minutes completion gate, are an added digital conservatism - not parameters specified by the randomised controlled trials.

Step 22. Re-test on cadence (Stage 4)

Re-test the HRt on a graded test **weekly** in the randomised controlled trials; **1-2 weekly** clinically.

Role: patient executes; clinician monitors and approves progression. **Stopping criteria:** red flag at any point; repeated provocation -> regress or refer.

Stage 5: Graded return**Step 23. Track the objective recovery curve (Stage 5)**

Serial re-testing tracks the threshold rising toward, or reaching, no-provocation. The BCTT is reliable as a repeated measure, which is what licenses serial testing as a recovery trajectory.

Step 24. Surface the clearance signal (Stage 5)

A re-test returning **no-intolerance** - voluntary exhaustion, RPE above 17, without provoking symptoms - surfaces automatically as a clinician clearance-review flag.

Step 25. Integrate with the consensus return strategy (Stage 5)

Feed clearance to progress aerobic load into the consensus graded Return-to-Sport and Return-to-Learn strategy: symptom-limited stepwise progression through activity stages with the same 2-point tolerance rule.

SAFETY - scope limit. This remains a clinician-led decision. The digital workflow supplies the objective aerobic-tolerance evidence; it does not, by itself, clear an athlete for contact.

Expected results and troubleshooting**Decision rules at a glance****Red flag at screen or mid-test**

SCAT6 emergency list positive - Halt, no HRt, urgent medical review

Symptom rise of 3 pts or more above rest during graded test

Diagnostic provocation criterion - Record HRt at that stage; prescribe band

RPE above 17 / 90%+ age-predicted max, no rise of 3 pts or more (initial)

No exercise-intolerance phenotype - Redirect workup (cervical/vestibular/mood)

RPE above 17, no rise of 3 pts or more (re-test)

Recovered tolerance - Flag clearance review; graded return

HRt below 135 bpm AND delta-HR 50 bpm or less

Prognostic flag (recorded, not a dose input) - Surface likely-prolonged-recovery signal

Symptom rise of 2 pts or more during a home session

Within-session TRAINING stop rule - Stop session; log flare

Repeated recent provocation

Reduce, don't rest - Regress ceiling, continue daily at reduced intensity; consider re-test or referral

Run of clean verified sessions

Tolerance improving (verified sessions only) - Advance ceiling +5-10 bpm, capped at measured HRt

Band reaches measured HRt

Ceiling cap - Prompt re-test; never advance past the measurement

Known limitations and evidence gaps

1. **There is no validated severity-adjusted starting dose.** The percentage-of-HRt band is the only individualisation the evidence supports. A more cautious starting point for a highly symptomatic patient is clinician judgement, not an evidence-derived default.
2. **The trials specify no next-day symptom-contingent rule.** This workflow records a next-day flare flag and lets the clinician act on it; the reduce-don't-rest response derives from the general management principle, not a trial-specified next-day algorithm.
3. **There is no published protocol for a large or delayed flare.** Management of a substantial or late-onset symptom spike - as distinct from the transient within-session worsening of 2 points or less the consensus tolerates - is not specified in the literature.

Route such events to clinician review rather than encoding an unvalidated automated response.

4. **There is no specified washout between the diagnostic graded test and programme start.** The evidence does not define an interval; this is left to clinician timing.
-

Declarations

Conflict of interest. The author is the founder and director of Concussion Education Australia Pty Ltd, which develops and commercialises software implementing this workflow, and which sells concussion education to clinicians. This protocol describes a workflow derived from published evidence; no efficacy claim for any commercial product is made or implied.

Ethics. No ethics approval was required. This protocol describes a workflow built on published evidence; no human participant data were collected.

Data availability. No datasets were generated or analysed. All parameters are drawn from the cited published literature.

Funding. None.

Key references

1. Patricios JS, Schneider KJ, Dvorak J, et al. Consensus statement on concussion in sport: the 6th International Conference on Concussion in Sport - Amsterdam, October 2022. *Br J Sports Med.* 2023;57(11):695-711.
2. Leddy JJ, Haider MN, Ellis MJ, et al. Early subthreshold aerobic exercise for sport-related concussion: a randomized clinical trial. *JAMA Pediatr.* 2019;173(4):319-325.
3. Leddy JJ, Master CL, Mannix R, et al. Early targeted heart rate aerobic exercise versus placebo stretching for sport-related concussion in adolescents: a randomised controlled trial. *Lancet Child Adolesc Health.* 2021;5(11):792-799.
4. Leddy JJ, Willer B. Use of graded exercise testing in concussion and return-to-activity management. *Curr Sports Med Rep.* 2013;12(6):370-376.
5. Leddy JJ, Haider MN, Hinds AL, Willer B. Practical management: prescribing sub-symptom threshold aerobic exercise for sport-related concussion in the outpatient setting. *Clin J Sport Med.* 2021;31(2):e89-e94.
6. Haider MN, Leddy JJ, et al. The predictive capacity of the Buffalo Concussion Treadmill Test after sport-related concussion in adolescents. *Front Neurol.* 2019;10:395.
7. Haider MN, Leddy JJ, Willer B, et al. Exercise for sport-related concussion and persistent postconcussive symptoms. *Sports Health.* 2021.
8. Leddy JJ, Baker JG, Kozlowski K, Bisson L, Willer B. Reliability of a graded exercise test for assessing recovery from concussion. *Clin J Sport Med.* 2011;21(2):89-94.
9. Leddy JJ, et al. Practical management: a standardized aerobic exercise program for adolescents with concussion in the absence of graded exercise testing. *Clin J Sport Med.* 2023.

10. Leddy JJ, Haider MN, Ellis M, Willer BS. Exercise is medicine for concussion. *Curr Sports Med Rep*. 2018;17(8):262-270.
11. Willer BS, Haider MN, Bezherano I, et al. Comparison of rest to aerobic exercise and placebo-like treatment of acute sport-related concussion in male and female adolescents. *Arch Phys Med Rehabil*. 2019;100(12):2267-2275.